
EC ENGR C143A/C243A Final Report

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Abstract

In this project, we investigate neural speech decoding from intracranial recordings using the Brain-to-Text Benchmark '24 dataset. Starting from the provided GRU-based baseline decoder, we systematically explore alternative data transformations, augmentation strategies, and model designs, including log-transformed neural features before normalization, varied noise and baseline-shift augmentations, and attention-based sequence models. Our final approach replaces the recurrent backbone with a Conformer architecture operating on non-overlapping temporal patches of the neural signal and equipped with causal self-attention and relative positional encoding to enable streaming decoding while preserving temporal structure. This Conformer-based model achieves a Phoneme Error Rate (PER) of 16.48% on the validation set, representing a substantial improvement over the baseline GRU decoder and underscoring the importance of jointly optimizing preprocessing, augmentation, and architecture for accurate neural speech decoding. For grading, our code and modelweights can be accessed until next quarter, through <https://github.com/WHYeats/ECEC143-243-Neural-Speech-Decoder> and https://drive.google.com/drive/folders/1gJrEZDsMa01PtgPK4-rhXLbdVTrtf5k7?usp=drive_link, respectively.

1 Introduction

Neural speech decoding aims to reconstruct text from brain activity and is a central problem in brain-computer interfaces for communication[1]. The neural dataset used in this project comes from the Brain-to-Text Benchmark '24[2], a competition designed to encourage new machine learning methods for decoding speech from intracranial recordings. Along with the dataset, the organizers released a baseline gated recurrent unit (GRU) model, and the main goal of the benchmark is to improve neural speech decoding performance beyond this baseline. The task is to predict a sequence of linguistic units for each sentence given its associated neural activity. In this benchmark, text labels are converted to phonemes, which more closely match the speech sounds the participant attempts to produce. Because there is no ground-truth alignment between phoneme sequences and neural time bins, models are trained using Connectionist Temporal Classification (CTC) loss. Performance is primarily evaluated using the Phoneme Error Rate (PER), which is computed from the number of substitutions, deletions, and insertions relative to the reference phoneme sequence.

1.1 Neural Dataset

The Brain-to-Text '24 benchmark dataset consists of intracranial microelectrode array (MEA) recordings from the ventral premotor cortex (area 6v) of a single participant with ALS. Neural activity is collected from two 64-channel arrays (ventral and dorsal), producing 256 features per time bin derived from spike band power and threshold crossings. During each trial, the participant is shown a sentence and attempts to speak it at the “go” cue, and all decoding analyses are performed on neural activity during this go phase. The signals are provided in 20 ms bins (50 Hz) and z-scored within blocks of 20–50 sentences. The competition dataset[1] includes 10,880 total sentences, organized into fixed train, validation, and test splits. The training set contains 8,800 sentences, the validation set contains 880 sentences, and the test set contains 1,200 sentences. Training and validation recordings span 24 days over approximately four months, while the test set is drawn from 15 of these days. Because text labels for the test set are not publicly available, test performance can only be evaluated by submitting predictions to the competition server.

1.2 Evaluation Metrics and CTC

The goal of this benchmark is to predict a phoneme sequence for each sentence from the corresponding intracranial neural activity. Model performance is primarily evaluated using the Phoneme Error Rate (PER or CER), defined as

$$\text{PER} = \frac{S + D + I}{N}, \quad (1)$$

where S , D , and I are the numbers of substitutions, deletions, and insertions, and N is the total number of target symbols. Although it is also possible to compute Word Error Rate (WER) by converting decoded phonemes into words using a weighted finite-state transducer (WFST)–based decoder, this is not required in the project due to its additional computational cost; instead, the main focus is on improving validation PER.

A key challenge in this setting is that there is no ground-truth temporal alignment between the neural time series and the phoneme sequences. To handle this, models are trained with Connectionist Temporal Classification (CTC) loss[3], which is specifically designed for sequence labeling with unsegmented inputs. CTC sums over all valid alignments consistent with the target phoneme sequence, allowing the model to learn both the timing and identity of phonemes jointly from data. The CTC loss is already implemented in the provided codebase and serves as the standard training objective for all models in this project.

1.3 Baseline Gated Recurrent Unit Model

Along with the dataset, the benchmark provides a baseline gated recurrent unit (GRU) model[4]. The input features first pass through a day-specific linear layer and nonlinearity to account for day-to-day changes in neural activity. The transformed features are then fed into a 5-layer GRU that processes overlapping temporal windows of length T_{in} and produces phoneme logits every T_{out} steps. For optimal performance, the baseline algorithm sets $T_{in} = 32$ and $T_{out} = 4$, resulting in an 87.5% overlap between consecutive inputs. The model is trained with the Adam optimizer and CTC Loss. During training, white noise and baseline shift augmentations are added to the neural activity, followed by causal Gaussian smoothing. Training is performed for 10,000 batches. The model is trained with the Adam optimizer and CTC loss, using white noise and baseline-shift augmentations followed by causal Gaussian smoothing.

2 Methods

2.1 Data Preprocessing

Our preprocessing pipeline follows the structure of the released dataset but includes two additional steps applied before model training.

Block-wise z-score normalization The dataset is organized into recording blocks, each containing 20–50 consecutive sentences. Following the official preprocessing design, we perform block-wise

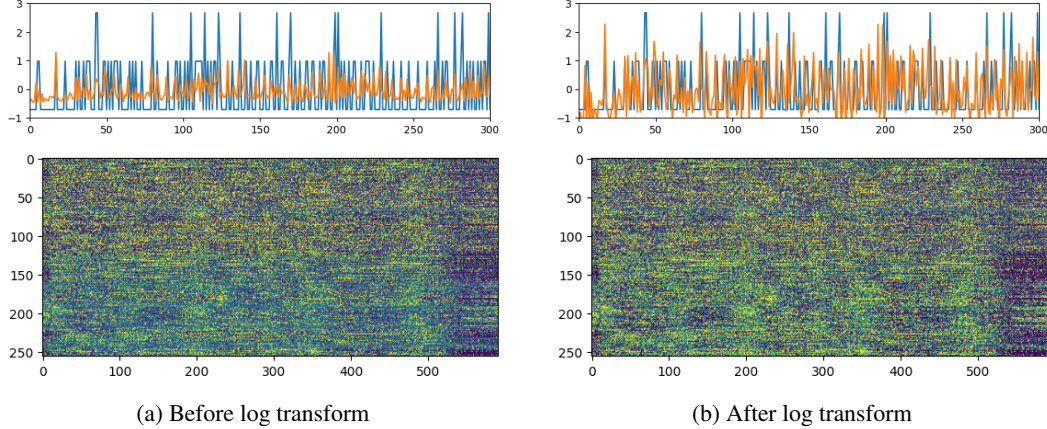


Figure 1: Effect of the log transform on neural features for a single training sentence (`trainDatasets[0]['sentenceDat'][10]`). In each subfigure, the top panel shows feature 0 (threshold crossings, blue) and feature 128 (spike-band power, orange), and the bottom panel shows the full 256-dimensional feature matrix; (a) before and (b) after applying the log transform to spike-band power.

z-score normalization: for each block, we concatenate all feature sequences in that block to compute a block-level mean and standard deviation. Each trial within the block is then normalized as

$$\tilde{x} = \frac{x - \mu_{\text{block}}}{\sigma_{\text{block}} + 10^{-8}}.$$

Log transform For each trial, we first extract 256 neural features consisting of 128 threshold-crossing channels and 128 spike-band power channels. As shown in our data-loading code, we apply a log transform to the spike-power features using

$$\log1p(x) = \log(1 + x),$$

which stabilizes the distribution and reduces the dynamic range of power measurements. This transformation is applied only to the spike-power portion of the feature vector (channels 128–255), consistent with prior work in neural signal normalization[5]. The effect is shown in Figure 1

2.2 Input framing and patching

The neural features are originally provided at a resolution of 20 ms per time step. We use different temporal framing strategies for the GRU baseline and for our Transformer-based models, reflecting the inductive biases of these architectures. For the GRU baseline, we follow the original sequence decoder and construct *overlapping* input windows: at each recurrent step the model receives a short temporal kernel of length `kernelLen` time steps, and the window is advanced by $T_{in} = 32 < T_{out} = 4$ time steps. This results in a high degree of overlap between successive windows, allowing the GRU to exploit fine-grained temporal continuity while still producing outputs at a higher effective rate than the raw 20 ms bins. In contrast, for our Transformer-based and Conformer models we use *non-overlapping* temporal patches, setting $T_{in} = 5$ equal to $T_{out} = 5$. Each input token therefore corresponds to a disjoint patch of neural activity. This design keeps the effective sequence length manageable for self-attention and matches the patch-based processing commonly used in Transformer-style models, while still providing sufficient temporal context within each patch.

$$N_{\text{patch}} = \left\lfloor \frac{T - T_{in}}{T_{out}} \right\rfloor + 1. \quad (2)$$

2.3 Augmentations

We investigate temporal masking as a form of data augmentation to improve robustness to missing or noisy neural activity. In this work, we consider two related strategies.

Algorithm 1 Temporal frame masking

Require: Feature sequence $\mathbf{X} \in \mathbb{R}^{L \times D}$, max mask length $L_{\max}$, number of masks N_{mask}

```
1:  $\tilde{\mathbf{X}} \leftarrow \mathbf{X}$ 
2: for  $n = 1$  to  $N_{\text{mask}}$  do
3:   Sample mask length  $L \sim \text{Uniform}(1, L_{\max})$ 
4:   Sample start index  $s \sim \text{Uniform}(1, T - L + 1)$ 
5:   for  $t = s$  to  $s + L - 1$  do
6:      $\tilde{\mathbf{x}}_t \leftarrow \mathbf{0}$  {mask frame  $t$ }
7:   end for
8: end for
9: return  $\tilde{\mathbf{X}}$ 
```

Temporal masking We first apply temporal masking at the frame level, where we randomly select contiguous time intervals and replace the corresponding feature vectors with zeros. This encourages the model to rely on a broader temporal context rather than overfitting to specific local segments. The algorithm is realized in Algorithm 1. We have different settings for GRU and Transformer based method. However, because our input is later grouped into temporal patches (time bins), frame-level masks can become diluted when multiple frames are averaged or concatenated into a single patch. In other words, temporal masking sets the features within the chosen time interval to zero while leaving the remaining frames unchanged.

Patch masking To address this issue, we also explore *patch masking* (or time-bin masking), which is used in [5], in which we operate directly on the patched input sequence. We set the mask ratio to $r = 0.075$ and the number of masking operations to $N_{\text{mask}} = 20$ across all experiments, which results in masking approximately 53% of patches on average.

In our experiments, we compare these two masking strategies and analyze their impact on validation performance.

2.4 Model architectures

GRU baseline and variants. We begin by revisiting the provided GRU-based baseline and exploring several architectural and optimization variants. The baseline model consists of a 5-layer GRU with day-specific linear input layers, trained with CTC loss and the augmentations provided in the benchmark code. Building on this configuration, we tune key hyperparameters such as the hidden size, number of layers, dropout rate, and training schedule while keeping the overall decoding setup (CTC objective, input features, and evaluation metric) fixed for a fair comparison. We also investigate the effect of model directionality by comparing unidirectional and bidirectional GRUs. The unidirectional GRU satisfies the streaming constraint of the task, whereas the bidirectional GRU can use both past and future context and thus serves as a non-causal reference with typically higher validation performance but no real-time guarantee.

Basic Transformer. To move beyond recurrent models, we consider a Transformer[6] encoder with 5 layers, matching the depth of the baseline GRU to enable more direct comparison as it is used in [5]. Instead of standard sinusoidal positional encodings, we adopt learnable relative positional encodings similar to those used in T5[7], allowing the attention mechanism to model relative offsets between time steps.

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}} + \mathbf{B} + \mathbf{M}\right) \mathbf{V}$$

$\mathbf{B}$ for relative position and $\mathbf{M}$ for causal mask. To improve training stability, we use a pre-layer normalization architecture[8], applying layer normalization before the self-attention and feed-forward blocks. We further compare two attention masking schemes: (i) a causal mask, where each time step can only attend to its past and current positions, and (ii) a full-context mask, where each time step can attend to the entire sequence. Under a fixed training budget (same number of batches), we find that the causal Transformer achieves better validation performance than the non-causal variant.

Conformer. Motivated by its strong performance in automatic speech recognition[9], we also adopt a Conformer encoder in Figure 2, which combines convolutional modules with self-attention to capture both local and global structure. In our setup, the Conformer operates on non-overlapping temporal patches of the neural features, and we again use learnable relative positional encodings, even though the convolutional module itself provides some positional sensitivity. As with the basic Transformer, we evaluate both causal and non-causal attention for the Conformer. We observe that causal masking performs at least as well as the non-causal variant at the same number of training batches, while additionally satisfying the real-time decoding constraint imposed by the task.



Figure 2: Conformer architecture[9]

3 Results

All results in this section are reported as the best validation performance obtained for a single random seed (seed = 0). A more extensive multi-seed evaluation is left for future work.

3.1 Comparison of finetuning on GRU

Table 1 summarizes an ablation study of our GRU finetuning procedure. Starting from the baseline GRU with a validation CER of 22.28%, we first apply a log transform to the spike-power features, which yields a small but consistent improvement. We then reduce the initial learning rate and extend training by an additional 3k batches, resulting in a larger drop in CER. Next, we adopt an Adam optimizer with a smaller ϵ and a learning-rate schedule, which further improves validation performance. Continuing training for another 3k batches provides an additional gain, and finally adding temporal masking leads to our best GRU result with a validation CER of 18.01%. Thus, the final performance is obtained by incrementally combining log-transformed features, tuned optimization hyperparameters, extended training, and temporal masking on top of the original baseline GRU.

The choice of temporal masking is also tied to our input framing strategy in (2.3). For the GRU baseline we use highly overlapping time bins, with a window length T_{in} and stride T_{out} such that more than 85% of consecutive windows overlap. Temporal masking is applied at the original frame level, so masked frames are still mixed with many unmasked frames across overlapping windows, which dilutes its effect on the effective GRU inputs and leads to only a modest CER gain. In contrast, for the Transformer-based and Conformer models in (2.4) we use non-overlapping temporal patches. In this setting, patch (time-bin) masking removes entire tokens from the input sequence and thus has a much stronger regularization effect than frame-level masking, making patch masking a more natural choice for Transformer-style architectures.

Table 1: Ablation study on GRU training configurations on the validation set

Configuration	Val CER (%)
Baseline (default)	22.28
+ log transform of spike power	22.19
+ lower initial learning rate and +3k batches	20.44
+ Adam with smaller $\epsilon = 1e-5$ and lr schedule from $1e-4$ to $1e-5$	18.70
+ further +3k batches (fine-tuning)	18.36
+ temporal masking	18.01

3.2 Comparison of Temporal and Patch Mask on Transformer-based model

All Transformer experiments in this subsection are trained for 10,000 updates with a batch size of 64, using a 5-layer Transformer encoder with sinusoidal positional embeddings. For temporal

Table 2: Comparison of temporal masking and patch masking on the Transformer encoder. Here T denotes the maximum sequence length (1200 frames).

Configuration	Masking strategy	Val CER (%)
$N_{\text{mask}} = 5, L_{\text{max}} = 0.025T$	Temporal masking	22.23
$N_{\text{mask}} = 5, r = 0.025$	Patch masking	22.31
$N_{\text{mask}} = 20, L_{\text{max}} = 0.075T$	Temporal masking	21.27
$N_{\text{mask}} = 20, r = 0.075$	Patch masking	20.88

masking, we mask N_{mask} contiguous frame segments with maximum length $L_{\text{max}} = 0.05 \times 1200$ or 0.075×1200 frames per sequence. For patch masking, we operate on the patched input sequence with the same N_{mask} and mask ratio r . Overall, Table 2 shows that temporal and patch masking behave similarly when the masking budget is small ($N_{\text{mask}} = 5$ and $L_{\text{max}}, r = 0.05$), with only minor differences in validation CER. However, when we increase the masking strength ($N_{\text{mask}} = 20$ and $L_{\text{max}}, r = 0.075$), patch (time-bin) masking yields a more stable improvement and achieves lower CER than temporal masking. Based on these observations, we adopt patch masking as our default augmentation strategy for Transformer-based attention models in the subsequent experiments.

3.3 Effect of learnable relative position and causal masking

We evaluate the impact of learnable relative positional encoding on both Conformer and Transformer architectures under non-causal and causal attention in (2.4). On average, replacing sinusoidal positional encodings with learnable relative encodings reduces the validation CER by about 1.85% for the Conformer and about 1.12% for the Transformer, indicating that relative position provides a substantially stronger positional signal than sinusoidal encodings in our setting in table 3. Interestingly, we also observe that restricting self-attention to be causal (only attending to past and current positions) often yields better validation CER than allowing full-context attention with the same architecture and training budget. This effect may be related to the structure of the neural signals and the CTC objective: both favor a predominantly forward, monotonic alignment between neural activity and the target sequence. As a result, enforcing a causal inductive bias in self-attention can be more compatible with the underlying decoding problem than fully non-causal attention, despite the latter having access to strictly more context.

Table 3: Effect of learnable relative positional encoding on Conformer and Transformer.

Model configuration	Sinusoidal Val CER (%)	Relative Val CER (%)
Conformer (non-causal)	18.39	16.48
Conformer (causal)	18.64	16.91
Transformer (non-causal)	19.91	18.73
Transformer (causal)	19.85	18.94

3.4 Comparison of different architecture

Table 4: Best model performance on the validation set.

Model	Val CER (%)	Training Epochs
Baseline GRU	22.28	10000
GRU	18.01	13000
Transformer	18.73	15000
Conformer	16.48	15000

4 Discussion

Impact of Preprocessing and Model-Specific Augmentations. The experimental results showcased several factors that influenced neural speech decoding performance on the Brain-to-Text Benchmark '24 dataset. Signal preprocessing techniques such as log-transforming spike-band power features, adjusting learning-rate schedules, extending training were able to reduce the validation CER from 22.28% to 18.01%. Relatively, temporal masking produced limited effect in the GRU as the highly overlapping recurrent windows diluted the masked segments. In contrast, augmentation had a stronger effect in attention-based architecture. Since Transformer and Conformer models operate on non-overlapping temporal patches, patch-level masking will remove entire tokens. When masking approximately 7.5% of patches, validation CER improved. Frame-level temporal masking yields smaller and less stable gains under comparable masking budgets. This trend exemplifies how augmentation strategies must align with the tokenization structure and inductive biases of the model.

Role of Positional Encoding and Architectural Design. Our results suggest that architectural choices interact strongly with how temporal structure is represented in neural speech signals as it is suggested in Table 4. Compared with the GRU baseline, Transformer-style models benefit from greater parallelism and a stronger capacity to model long-range dependencies, while the Conformer further improves performance by combining self-attention with convolutional inductive biases that emphasize local temporal patterns. This hybrid design appears well matched to intracortical signals that contain both short-timescale structure and broader contextual dependencies, allowing the model to capture fine-grained dynamics without sacrificing global sequence modeling. In this setting, learnable relative positional encoding is particularly important: rather than anchoring representations to absolute time indices, relative encodings emphasize distances and local order, which may be more robust to session variability and the non-stationary nature of neural features. We also observe that removing the causal mask degrades performance. This seems a little bit not logical since causal mask compromises the attention-model’s ability to understand in a whole context. A plausible explanation is that, although bidirectional context can be helpful in some sequence tasks, the neural-to-phoneme mapping under CTC relies on stable monotonic alignments. Allowing unrestricted future context can conflict with the monotonic alignment bias of CTC. In practice, non-causal attention may let the model rely on future frames to explain current labels, reducing the need to establish temporally precise evidence. With noisy and non-stationary neural signals, this shortcut can blur timing cues and hurt generalization. By contrast, a causal mask restricts future lookahead and effectively regularizes the model toward using locally grounded, time-consistent features, which can stabilize CTC-style alignments. Together, these results suggest that decoding quality depends not only on model capacity, but also on matching architectural priors—locality, relative timing, and controlled directionality—to the temporal structure of neural signals and the alignment assumptions of CTC.

Limitations and Future Directions. Several limitations remain. First, a single random seed was used for all experiments, providing limited information about variance across initialization and sampling conditions. Future work should report multi-seed results with confidence intervals to improve reproducibility and strengthen the reliability of observed ablation trends. Second, although we aimed to keep comparisons consistent, architecture changes may implicitly alter effective optimization dynamics and compute efficiency. A more rigorous evaluation would control for training budget (e.g., matched update steps, wall-clock time, or FLOPs) and report sensitivity to key hyperparameters.

Third, the study emphasized supervised training only, whereas self-supervised or masked-prediction pretraining may improve sample efficiency and robustness. Finally, while the conformer achieved relatively strong performance, the current approach still relies on session- or day-specific handling, suggesting that neural non-stationarity is not yet fully addressed. Incorporating domain adaptation, continual learning, or test-time calibration may further improve cross-day generalization.

5 Conclusion

In this project, we systematically improved neural speech decoding performance to the Brain-to-Text Benchmark '24 by optimizing preprocessing, augmentation, and model architecture. In the GRU baseline, we improved performance from log-transformed neural features, refined optimization schedules, and longer training. In the attention-based architectures, we found that patch-level masking,

causal self-attention, and learnable relative positional encodings enhanced the decoding accuracy. The conformer-based model achieved a validation CER of 16.48%, outperforming GRU and transformer.

Overall, the results demonstrate that improvements in preprocessing, augmentation, and architecture all contributed to decoding quality. Significant opportunities remain, particularly in scaling model capacity under controlled compute settings, incorporating self-supervised objectives, and explicitly addressing neural signal non-stationarity. Continued progress along these directions may yield further advances in decoding accuracy and help translate intracortical speech BCIs toward clinically viable communication systems.

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